Critical Stations and Live Disruptions in the New York City Subway

Master Data and Intelligence for Smart Systems Data Processing and Analytics Project

Made By: Oussama Sbaa / Mazen Morsy / Abdullah Hameed

GitHub repository link : https://github.com/oussama-sbaa1/DPA_Project_NYC_SUBWAY

1/ Domain and Questions

A. Domain: public transport, the New York City subway: critical stations and live disruptions

B. Main question: Which subway stations are critical for the network, and are live disruptions happening in those stations or not?

C. Sub-questions for each part:

Part 1 · Volume (Spark SQL): What is the usual ridership in each subway station by hour and day of the week, and by how much does the weather change it?

Part 2 · Velocity (Structured Streaming): Right now, which stations have live alerts for disruptions, and how busy are those stations normally at this hour?

Part 3 · Variety (GraphFrames): Which stations are the most central in the New York subway network, and are those central stations also the busiest ones?

2/Data sources for each part: links, sizes and licences

Part	Dataset	Link	What we take · size	Licence
1	MTA Subway Hourly Ridership: 2020–2024	data.ny.gov/d/wujg-7c2s	Jul–Dec 2024 **14.0 M rows · 2.43 GB**	OPEN-NY
1	MTA Subway Hourly Ridership: Beginning 2025	data.ny.gov/d/5wq4-mkjj	Jan–Jun 2025 **15.4 M rows · 2.65 GB**	OPEN-NY
1	NOAA Local Climatological Data v2, Central Park (USW00094728)	ncei.noaa.gov (LCD v2 API)	Hourly, Jul 2024–Jun 2025 **11,765 rows · 5.7 MB**	U.S. public domain, cite NOAA
1·2·3	MTA Subway Stations (links complexes, stop IDs and GTFS stops)	data.ny.gov/d/39hk-dx4f	Whole table **496 rows · 69 KB**	OPEN-NY
2	MTA live subway alerts (GTFS-realtime, JSON), no API key	api.mta.info → camsys/subway-alerts.json	1 read per minute **~1 MB per read ≈ 1.4 GB/day**	MTA data feed terms
2	MTA Service Alerts: Beginning April 2020 (backup, replayed as a stream)	data.ny.gov/d/7kct-peq7	Subway alerts **291,401 rows · 75 MB**	OPEN-NY
3	MTA static GTFS, subway (regular network)	rrgtfsfeeds.s3.amazonaws.com/gtfs_subway.zip	stops, stop_times, transfers **5.6 MB zip · 43 MB unzipped**	MTA data feed terms

Part 1 volume: 29.4 M rows · 5.08 GB of CSV for 12 months .

Parts 2 and 3 volume: Also reuse our Part 1 table of normal ridership per station, weekday and hour.

N.B: Raw files on the original database should be kept untouched According to the licensing rules for MTA database regulations, therefore we created on own copies to manipulate.

3/ Main risks and proposed solutions

- **Rate limits:** data.ny.gov downloads are slow the MTA can block us if we read the live feed too often → Solution proposed: We will download month by month with retries. This will enable us to get the data for each month alone.
- **Missing periods:** live data that is not recorded is lost forever → Solution proposed: We will start recording early and save every snapshot so it can be replayed.
- **Few live disruptions:** only a handful of alerts are active at the exact moment we look at the live data. (3 channels out of 15 that dispay the real live data)
- **Licensing:** the MTA says "do not modify the data" → Solution proposed: keep the raw files untouched and clean into separate tables.
- **Joining the sources:** IDs, time zones and stop codes do not match across sources → Solution proposed: convert them into one agreed form and check with row counts.